The earlier large-scale results still count

They support a separate, physically meaningful sensing benchmark—not a Quiroga reproduction.

Qubits / run	Gaussian noise σ	Updates	Factor fidelity	Normalized TP
8 / continuation	0.05	200,000	99.1393%	0.01066
8 / sampling seed 1	0.05	200,000	99.2163%	0.01068
8 / sampling seed 2	0.05	200,000	98.9730%	0.01065
8 / sampling seed 3	0.05	200,000	98.9189%	0.01065
10 / reduced noise	0.0015625	57,000	99.0045%	0.01595
10 / original noise	0.05	100,000	0.004015%	0.32522

Eight qubits: about 99% factor fidelity is repeatable across the tested sampling streams. These are the SAME channel, observations and initialization—not four independent channel recoveries.

Ten qubits: the high-fidelity endpoint used 32× lower noise. Recovery at σ = 0.05 was not established. The later cancelled continuation reached only 0.018935% at its last inspected checkpoint (not a terminal result).

TP is the residual norm divided by $\sqrt{(2^n)}$. Every endpoint here is above the newer 0.001 TP threshold; high normalized factor overlap does not certify an exactly trace-preserving channel.

What changed was mainly the sensing experiment

Both benchmark families use rank-one Haar-unitary targets; neither is a higher-rank channel study.

Feature	Earlier local-Pauli benchmark	Current Quiroga-style benchmark
Input states	Products of $\|0\rangle$, $\|1\rangle$, $\|+\rangle$, $\|+i\rangle$	Basis states + global pairwise superpositions
Readout	Local X/Y/Z settings and outcomes	One generalized 2d-outcome POVM
Full virtual row count	24^n (≈110 billion at n = 8)	2×8^n (≈33.6 million at n = 8)
Highlighted data	Additive Gaussian observation noise	Latest calibration is noiseless
Interpretation	High-fidelity noisy reconstruction	Strict fidelity / RMSE / TP gate

The old inputs and projective measurements are physical, not arbitrary random sensing matrices. The Gaussian corruption is a synthetic observation model, not finite-shot counts. It does not turn the target into a noisy or mixed quantum channel.

There is also a convention difference: for the same generated unitary V, the legacy probabilities correspond to conj(V), while the new backend uses V. Both are valid unitary channels with the same Haar distribution, but same-seed runs are not identical physical instances.

Consequently, the old fidelity curves and GPU engineering remain useful. Their numerical tuning, convergence and comparison claims cannot simply be transferred to the new probe/POVM geometry.

How to present the older work to coauthors

Keep two result tracks: large-scale local-Pauli recovery and matched Quiroga-style comparisons.

Suggested headline

“FRAMES achieved approximately 99% factor fidelity on noisy eight-qubit local-Pauli QPT, and at ten qubits with reduced noise. Trace preservation remained approximate; these are separate from our Quiroga-style tests.”

Baseline evidence: the paired eight-qubit projected momentum-SGD runs remained near 8.73×10^{-6} fidelity, versus 0.9892–0.9922 for FRAMES, at 200,000 updates. Minibatches were matched; this is evidence about the tested baseline settings, not a general superiority theorem or a published Quiroga reproduction.

Engineering evidence survives: on-demand fixed observations, structured operators, bounded batch workspace and restartable GPU runs enabled these large instances without storing the full observation table. Virtual row counts are not counts of experimentally acquired measurements.

These noisy runs were not designed around today’s noiseless joint gate. Present them as high-fidelity reconstructions with approximate TP, not exact-recovery passes; retain their original objectives.

Evidence / verification scope

8q: archived NumPy full-factor verification + locally rechecked result hashes (continuation and all six paired runs).
10q: preserved full-factor verification record and completed-run table; native 10q factor not reloaded for this addendum.
Design: docs/qpt_quiroga_sensing.md. Preservation: higher-qubit audit of 4 October.
Exact file paths, input checksums and numerical records: legacy_addendum_evidence.json beside this PDF.